

IoT-Based Greenhouse System with Cloud Data Logging and AI Agent Consultation

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For the Bachelor of Information Technology (Hons) in
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DECLARATION

I declare that this report entitles “IoT-Based Greenhouse System with Cloud Data Logging and AI Agent Consultation” is my original work and all references have been citedadequately as required by the University.

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APPROVAL PAGE

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ABSTRACT

The intelligent IoT greenhouse platform presented in this study combines cloud storage, generative AI, and an ESP32 edge network for precision farming. The system broadcasts live streams over MQTT while continuously monitoring soil moisture, temperature, and humidity. A Node-RED pipeline combines various telemetry feeds and performs real-time logging to a Firebase Realtime Database to guarantee data retention. At the same time, an ElevenLabs Conversational AI agent receives real-time metrics via a context-aware data engine. In addition to human control overrides, this specialist agronomy model parses environmental circumstances to provide natural-language diagnostic health notifications straight to a web dashboard. The findings show that a highly scalable, data-driven framework for contemporary agricultural diagnostics may be created by combining lightweight edge telemetry with reliable cloud databases and conversational language models.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Agriculture is one of the common topics when it comes to modernization. As more technology rises, agriculture technology must keep up and adapt as it is a necessity to mankind. One of the best examples of the change of technology is the usage of sensors and machines for agriculture such as temperature sensors, moisture sensors, and more to keep the crop in the perfect condition.

Greenhouse is one of the methods of modern agriculture that is most popular. The idea behind greenhouse is to create an Ideal environment for a crop to grow. For example, creating enough humidity and temperature for a crop to grow healthy by having a fan that lets out the heat surrounding the crop. The issue with greenhouse is it still requires human and manual use making it not sustainable and inefficient.

Internet of things (IoT) is a technology that transfers data between sensors and actuators using microcontrollers. IoT is mostly embedded to household appliances such as smart tv or smart fan. With internet of things, data able to transfer from network to network seamlessly without any human interaction.

IoT-Based Smart Greenhouse System with Cloud Data Logging and AI Agent Consultation is an IoT project that will be implemented to Greenhouse so that the process of the greenhouse will be done automatically. By combining IoT with AI agent, the system can analyze data, make decisions, and automatically adjust the greenhouse environment to maintain ideal condition.

1.2 Problem Statement

Maintaining an ideal environment for crops is essential for maximizing yield and health in precision agriculture. Greenhouses are designed to optimize environmental settings such as temperature, humidity, and soil moisture. However, traditional farming setups heavily rely on visual observation and manual monitoring, which leads to inconsistent environmental management. Even when farmers utilize basic automated setups, they often operate blindly without real-time data logging or historical data retention, making long-term analytical tracking and trend prediction impossible.

Furthermore, traditional monitoring systems lack data intelligence and diagnostic accessibility. While basic sensor networks can display raw data values on local screens, they fail to interpret what those numbers mean for plant health. When environmental conditions fluctuate unexpectedly, non-technical farmers experience a delayed or inefficient response because standard systems display numbers without providing actionable, context-aware insights. This communication gap between raw telemetry data and human understanding frequently results in crop stress, slower plant growth, or sudden harvest failure.

Therefore, modern agriculture requires a centralized platform that can securely store historical telemetry and translate complex environmental metrics into accessible, real-time insights. By implementing an IoT telemetry network backed by a Firebase Realtime Database and an ElevenLabs conversational AI diagnostics agent, this project addresses the critical gap in data accessibility and retention. The system ensures continuous cloud data logging while transforming raw sensor variables into plain, natural-language expert advice, enabling farmers to make data-driven decisions seamlessly.

1.3 Project Objectives

- 1) To design an IoT based greenhouse system that monitors and collects the environment data such as temperature, humidity, soil moisture using sensor.
- 2) To implement cloud based ai agent capable of analyzing real time sensor data and making consultations.
- 3) To demonstrate the effectiveness of combining IoT and AI technologies in achieving precision agriculture and sustainable greenhouse management.

1.4 Project Scope

The scope of this project focuses on the development of a small-scale, intelligent IoT-based greenhouse monitoring and control platform designed to optimize data collection and diagnostic accessibility. The system is simulated and implemented using an ESP32 microcontroller to harvest real-time telemetry from environmental sensors, specifically capturing ambient temperature, humidity, and soil moisture levels.

Data transmission is routed via lightweight MQTT communication protocol blocks to a centralized Node-RED orchestration middleware layer. Rather than using the AI agent for direct physical actuation, the Node-RED backend coordinates a dual-layered data pipeline:

Cloud Data Logging Layer: Executes real-time data persistence by compiling synchronized sensor payloads and streaming them directly to a Firebase Realtime Database backend for historical tracking and analysis.

Conversational AI Core: Transforms raw telemetry streams into standard contextual parameters before passing them to an ElevenLabs Conversational AI Agent. This agent parses the environmental variables to output natural-language botanical consultation notices directly onto a centralized web-based user interface.

Environmental control functions are managed via dual-mode operational loops. In automatic mode, local threshold logic interlocks dictate actuator behavior, while a dashboard switch system allows users to seamlessly issue manual command overrides (e.g., controlling the exhaust fan).

This project is strictly limited to small-scale prototype simulation, environmental telemetry aggregation, cloud data logging, and generative AI diagnostics. It does not include visual camera attachments, computer vision modules for disease detection, or commercial-grade, large-scale hardware infrastructure.

1.5 Significance of project

This project contributes significantly to the field of smart agriculture by bridging the communication gap between raw edge sensor telemetry and actionable human understanding. By leveraging a specialized conversational AI agent, the platform translates complex, numeric data streams into friendly, natural-language diagnostic notices. This enables farmers—including non-technical operators—to understand exactly how their crops are faring and make swift, data-driven decisions without needing to manually interpret abstract sensor graphs.

From a data architecture perspective, the integration of a Firebase Realtime Database ensures structured historical data retention. This establishes a robust data pipeline that eliminates blind manual monitoring, allowing for future trend analysis and long-term environmental tracking.

Furthermore, the implementation of local automation coupled with a remote manual override dashboard reduces constant manual supervision, mitigates human error in climate regulation, and ensures resource optimization. Ultimately, this research provides a highly scalable, intuitive, and accessible framework for technology-driven precision farming and sustainable crop management.

CHAPTER 2: LITERATURE REVIEW

2.1 Internet of Things (IoT) in Agriculture

Internet of Things (IoT) is A network of connected devices that connect to and share data with the cloud and other IoT devices, IoT devices includes both digital and physical machines. This technology mostly implements on a goods that required sensors and software. In terms of agriculture, IoT is a perfect technology to implement because the use of the sensors and the actuator for data collection and control

The IoT has become a transformative technology in agriculture, enabling the automation of processes that were once executed manually. By integrating sensors, microcontrollers, and wireless communication, IoT systems continuously collect and transmit environmental data such as soil moisture, temperature, humidity, and light. These real-time data streams allow for smarter resource management, helping to reduce labor and minimize waste to improve yield and efficiency (Role of IoT Technology in Agriculture MDPI).

A study has discussed the use of IoT in agriculture. The study highlighted that IoT is a highly promising technology necessary for the industrialization and intensification driven by the growing demand for food quality and quantity. The key contribution was the IoT offers a lot of solution on the main issues in agriculture. The review's key contribution was its comprehensive analysis of relevant research across four main domains: IoT agricultural applications, sensors/devices, communication protocols, and network types. Furthermore, the study presented an IoT agriculture framework, discussed country policies for IoT-based agriculture, and outlined open issues and challenges to guide future research directions in the domain.

According to various studies, IoT deployments typically use a three-layer architecture: the perception layer (sensors), the network layer (communication), and the application layer (data processing and decision-making) (A Revisit of IoT Technologies for Monitoring and Control Strategies MDPI). This scalable, layered design is crucial for implementing comprehensive systems, such as those used in smart greenhouses, where sensors automate environmental control to maintain optimal crop conditions and enhance productivity (Revolutionizing Agriculture: IoT Empowered Smart Greenhouses sjpas.com). Ultimately, IoT serves as the foundational technology that enables real-time sensing and remote control, setting the groundwork for more advanced systems enhanced with artificial intelligence (AI).

2.2 Greenhouse Automation Systems

Greenhouse farming is a controlled environment where temperature, humidity, and lighting conditions can be regulated to obtain ideal plant growth. Traditional greenhouse systems rely heavily on manual monitoring the environment and adjust such as opening vents, switching on fans, or watering plants based on observation. This approach is time-consuming, inconsistent, and tend to have any human error.

To overcome the limitation of traditional greenhouse, research on IoT-based greenhouse automation systems has been conducted. The system that chosen to that has been studied are IoT-enabled Greenhouse Systems: Optimizing Plant Growth and Efficiency by (Manoharan, Swathi. 2024) and IoT Based Smart Greenhouse and Smart Farming by (Mishra, Rudra Kumar. 2023). These systems use sensors to collects data on environmental conditions and actuators such as fans, pumps, or servos to regulate them. Automated greenhouses help reduce labor work, improve consistency, and increase productivity compared to manual systems.

However, many existing greenhouse systems are limited with:

- Lack of intelligence
- No predictive capability
- Limited communication between multiple components

For this project, the use of AI agent in implemented. Due to that, a research paper on AI agent has been recorded to understand AI agent better and to choose the best approach of the AI agent in the smart greenhouse. Transforming Data Annotation with AI Agents: A Review of Architectures, Reasoning, Applications, and Impact (Karim, Md Monjurul. 2025) research paper has been read thoroughly for the purpose of understanding AI agent.

2.2.1 IoT-enabled Greenhouse Systems

The core research for the project is to create an automatic greenhouse using IoT technology. The research was conducted due to limitation of inaccuracy of traditional greenhouse. With IoT, real time update on crop data able to be received and automation able to achieve.

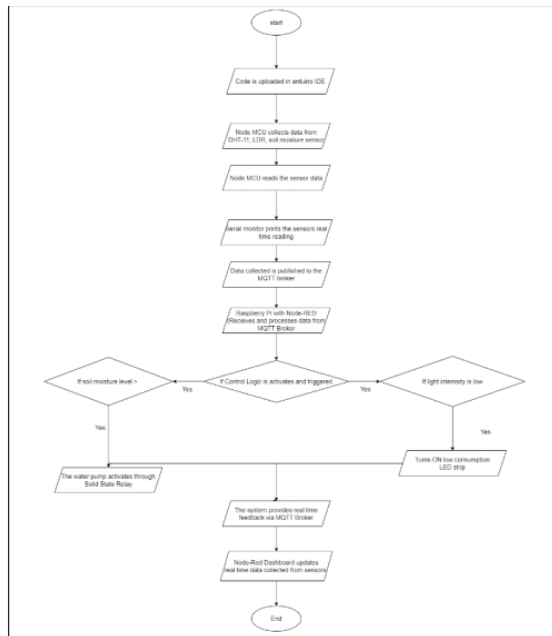


Diagram 1 - flowchart of IoT greenhouse system

The flowchart shows that the foundation flow of the system. The code was uploaded using Arduino IDE and uploaded to micro controller which in this system, node MCU. Next, the sensor will active and sending the data to MQTT through the node MCU to display to dashboard. While that, based on the code, the actuator will react from the data from node MCU. This is to create the perfect environment for the crop.

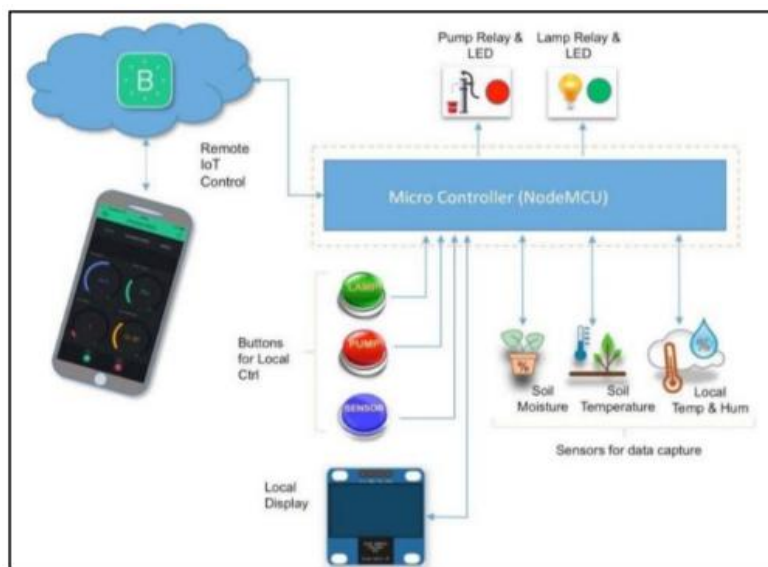


Diagram 2 - overview of IoT greenhouse.

2.2.2 IoT Based Smart Greenhouse and Smart Farming

This research is proposing a project that monitor and regulate for the optimal environment for cultivation. The project is expected to create an efficient greenhouse that reduces manual intervention and enhance growth. The project includes multiple sensors to record the data of the environment such as LDR sensor, DHT11, soil moisture, and flame sensor. It uses Arduino Uno as the microcontroller and nodeMCU for network communication with blynk integration. The project reports the collected sensor data and comparing growth of the plants.

2.2.3 Transforming Data Annotation with AI Agents: A Review of Architectures, Reasoning, Applications, and Impact

AI agents are increasingly combined with knowledge-retrieval mechanisms to improve decision accuracy. Retrieval-Augmented Generation (RAG) frameworks allow AI agents to access external knowledge repositories and generate informed responses grounded in verified information, thereby improving reliability and reducing incorrect or hallucinated outputs. This capability enhances decision-making in complex scenarios that require domain-specific reasoning, such as agriculture and environmental management.

Table 1 – types of ai agent

Architectural Paradigms	Strengths	Weaknesses
Single-Agent Sequential Pipeline	Simplicity, low integration overhead.	Susceptible to blind-spot bias; limited external validation.
Dual-Agent Reviewer Model	Explicit feedback loop, higher robustness.	Potential confirmation bias when agents share training data; higher compute cost.
Multi-Agent Collaboration	Role specialization, scalability across modalities, improved fault tolerance.	Communication protocol complexity; added latency.
Human-in-the-Loop (HITL) as an Agent	Guarantees expert oversight for high-risk domains (healthcare, finance).	Requires UI/UX design to keep interaction friction low.

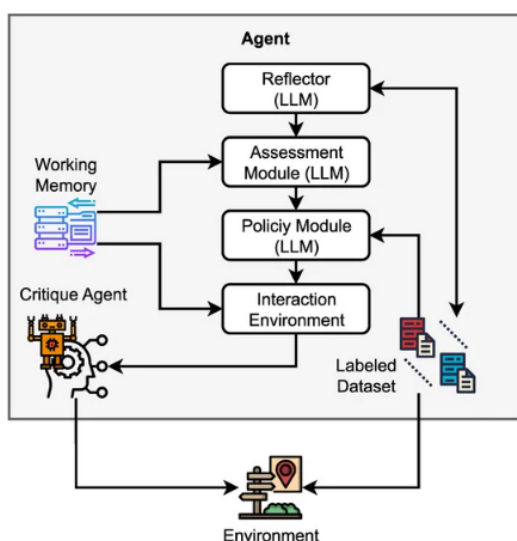


Diagram 3 - (the “agent-driven interaction loop”)

Diagram 3 is describing the internal feedback loop of an individual single-agent or dual-agent cycles that emphasise reflection, assessment, and policy modules.

The surveyed literature demonstrates a maturation from isolated LLM label generators to orchestrated multi-agent ecosystems that can actively select data, adapt guidelines, and enforce rigorous quality control. The most compelling visual aid for a review that emphasises this evolution is Figure 5, which clearly depicts the role-based agent ensemble, while Figure 2 is valuable for illustrating the internal reflective loop of a single agent. Together, these figures capture the architectural diversity and feedback-driven learning that define the next generation of AI-driven data annotation.

2.3 Artificial Intelligence in Agriculture

Therefore, the next step in smart greenhouses is the implementation of artificial intelligence. It can analyze data, learn patterns, and adjust greenhouse conditions more precisely compared to manual control.

Artificial intelligence (AI) is now widely used in every modern technology. AI is a technology that enables computers and machine learning to simulate human thinking to predict, learn, and making decisions. In agriculture, AI can support tasks such as plant sickness detection, irrigation, crop prediction, and environmental control. AI systems can process large amounts of data, recognize the patterns, and make predictions that help farmers improve decision making and reduce resource and/or wastage.

In summary, implementing AI greenhouse system can be used to:

- Analyze environmental trends
- Predict future temperature or humidity levels
- Recommend suitable irrigation or ventilation schedules
- Adjust actuators based on plant needs

Unlike the traditional greenhouse systems, AI considers multiple factors simultaneously and can learn from past data to improve accuracy over time. In conclusion, the use of AI in agriculture are effective in maintaining stable and efficient greenhouse conditions.

The project will implement the use of AI agent instead of creating an AI from scratch. The presence of AI agents allows environmental control decisions to be automated. Instead of following fixed rules, AI agents reason based on the data they receive, making them more adaptive, flexible, and capable of responding to complex scenarios

2.3.1 Workflow Automation Platforms (n8n)

AI is a capability of computer or machine to think like a human. AI agents however are systems that does the work or a certain task by designing the workflow. AI agent works with Large Language Model (LLMs) to produce the response based on the data from the user. In this process, the agent will learn to adapts and able to predict thus creating the ideal response.

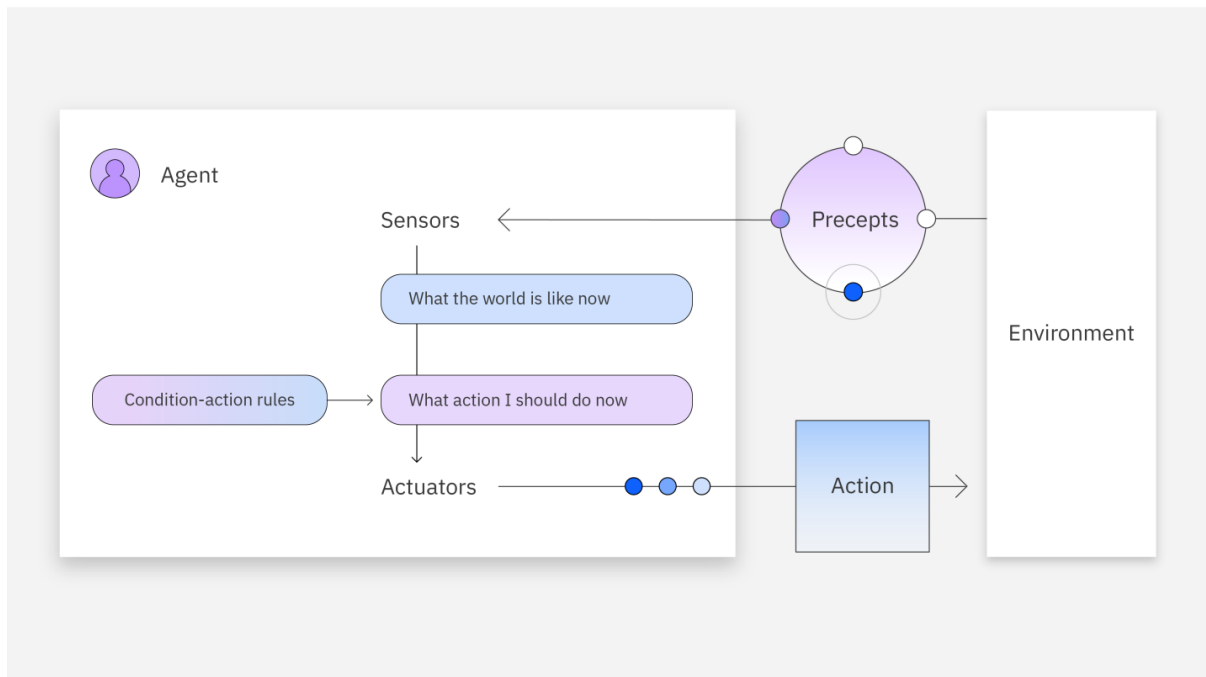


Diagram 4 - simple AI agent diagram

An n8n AI agent is an autonomous, intelligent workflow that uses LLMs to perform complex, multi-step tasks that are more dynamic and less rigid than traditional automations. Automation platforms such as n8n play an important role in connecting IoT devices with cloud applications, databases, and AI models. n8n is a low-code automation tool that allows users to design workflows visually without writing complex backend code. It operates similarly to Node-RED but with stronger cloud integration, modularity, and data processing capabilities.

In IoT applications, n8n is used for:

- Receiving data from microcontrollers via MQTT
- Processing and formatting data
- Sending data to AI agents or APIs
- Receiving AI responses and triggering actuator commands
- Creating dashboards or notification systems

n8n acts as the middleware that ensures smooth communication between the ESP32 devices and the AI agent. This makes the system scalable and easier to maintain. Instead of programming every data flow manually in the microcontroller, n8n handles the workflow logic and cloud communication, allowing the IoT device to stay lightweight.

2.3.2 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI technique that enhances decision-making by combining two abilities:

- Retrieving relevant information from a knowledge base
- Generating an informed response using that information

In simple terms, RAG allows AI to look up information before making a decision. This is extremely useful in agriculture, where many decisions depend on established guidelines such as:

- Ideal humidity for certain plant species
- Recommended irrigation cycles
- Appropriate light intensity levels
- Temperature thresholds for plant stress

By using RAG, the AI agent in this project can refer to agricultural knowledge (which you will provide as a mini database) before deciding how to control the greenhouse environment.

This makes the AI:

- More accurate
- More reliable
- More explainable (can justify decisions)
- RAG transforms the AI agent from a simple logic model into a more intelligent decision-support system.

2.4 Summary of Literature Review

The literature shows that IoT has significantly improved agriculture by enabling automatic data collection and environmental monitoring. However, most IoT-based greenhouse systems still depend on fixed rule-based controls, which limit adaptability and efficiency.

Table 2 - Comparison

Author	Reserch title	Focus area	Key contribution
Simo, A., Dzitac, S., Badea, G. E., Meianu, D. 2022	Smart Agriculture: IoT-based Greenhouse Monitoring System.	Real-time environmental monitoring within controlled agricultural spaces using edge microcontrollers.	Developed a baseline sensor-to-dashboard tracking architecture for temperature and humidity regulation.
Mishra, Rudra Kumar. 2023	IoT Based Smart Greenhouse and Smart Farming	monitor and regulate environmental conditions for optimal cultivation.	Collects data and display it
Manoharan, Swathi. 2024	IoT-enabled Greenhouse Systems: Optimizing Plant Growth and Efficiency	improving agriculture in controlled environments.	Sending the data through networks
Karim, Md Monjurul. 2025	Transforming Data Annotation with AI Agents: A Review of Architectures, Reasoning, Applications, and Impact	Designing and evaluating autonomous, LLM-powered AI agents that coordinate the entire data-annotation pipeline	An introduction on AI on and hardware

Artificial intelligence offers a solution by enabling smart decision-making based on environmental data. Workflow automation tools such as n8n further enhance system integration by simplifying data communication and cloud interactions. Finally, RAG strengthens AI reasoning by allowing the agent to reference agricultural knowledge before generating control decisions.

Based on these findings, this project combines IoT sensors, cloud AI agent, Node-RED automation, and RAG-enhanced decision-making to build an autonomous greenhouse system capable of maintaining optimal plant-growing conditions with minimal human intervention.

CHAPTER 3: METHODOLOGY

Methodology is a system of ways of doing, teaching, or studying something. In this project, methodology will be referred to documented process used to aid in developing it. This chapter will contain all the phases required to understand and to achieve the objectives projects.

3.1 introduction

The methodology will discuss on the requirements, design, and implementation. The chosen methodology will help to decide on the suitable method to implement and design the project as intended as stated in chapter 1. The chosen methodology can help to determine the best method for implementation, testing, documenting the project.

3.2 research structure

Research about AI agent, greenhouse requirements, a specific plant has been conducted and reported. Most are from the internet and article on environmental changes. Research comes from academic thesis on similar project. This has been done to justify the reasoning of the project and to refer the limitations and how to surpass it.

3.3 figure structure

3.2.1 Flowchart

The flowchart explaining the flow of this system. The system first starts with connecting to wifi. This is very crucial as the idea of the project is to connect to AI agent. Next after connection successful, the sensor now will start sensing the environment. In this project, the sensor will detect and collect the data from the moisture of the soil, temperature and humidity. Next, the data will be transfer to NODE-RED through the MQTT topic. AI AGENT will collects data from the NODE-RED. The AI agent will decide if the plant is in the ideal environment growth. If not, AI will send a consultation to the back to NODE-RED.

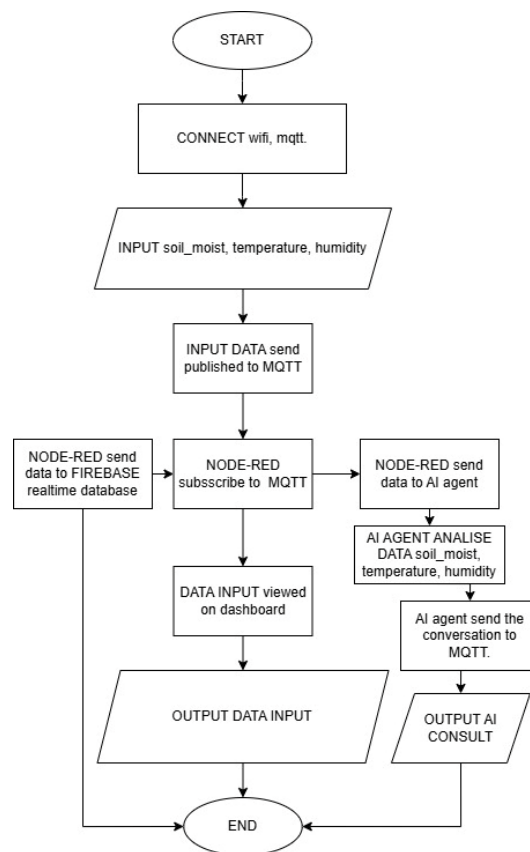


Diagram 5 - flowchart

3.3.2 Block diagram

The block diagram represents the data pipeline of the smart greenhouse. The flow moves from the edge sensors to the ESP32 microcontroller, which publishes to HiveMQ. Node-RED subscribes to this MQTT topic, displaying the live data on a user dashboard and splitting the data payload to Firebase (for logging) and the AI Agent (for consultation generation). Simultaneously, the microcontroller utilizes local threshold logic to actuate the cooling fan when temperatures exceed critical levels.

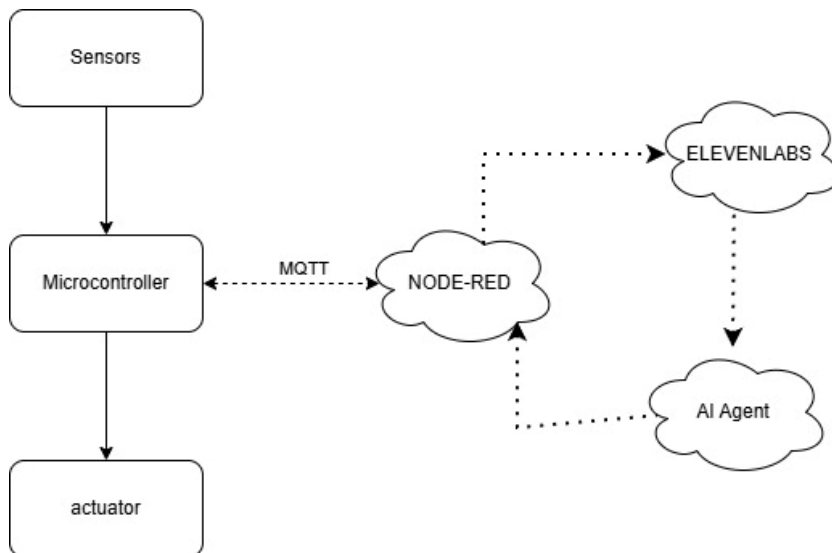


Diagram 6 - block diagram of the project.

3.3.3 Circuit Diagram

The circuit diagram shows how the systems hardware works.

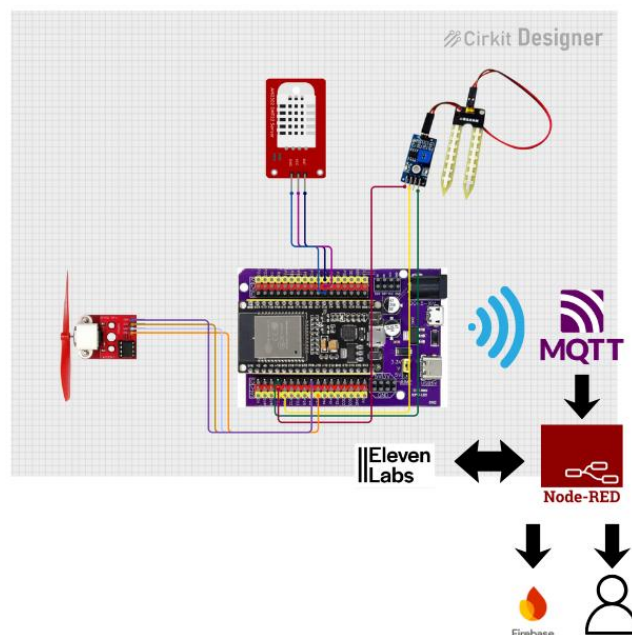


Diagram 7 - circuit diagram

a. Power Distribution

- **Primary Power:** The ESP32 is powered via a 5V USB connection.

b. Sensor Connections

- **DHT22 (Temp/Hum):** Connected to GPIO 4 with a 10kΩ pull-up resistor (if not built into the module) to ensure stable data transmission.
- **Soil Moisture Sensor:** The analog output (A0) is connected to an ADC-capable pin (e.g., GPIO 34) on the ESP32 to read the variable voltage representing moisture levels.
- **Relay Module (Fan):** Connected to a digital output pin (e.g., GPIO 27) using an active HIGH trigger to control the L9110 DC Exhaust Fan

In summary, the hardware will look like this:

Component	Pin Type	ESP32 Pin	Wiring Description	Component	Pin Type
DHT22	Digital Input	GPIO 4 (D4)	Signal pin. (VCC to 3.3V, GND to GND).	DHT22	Digital Input
Soil Moisture	Analog Input	GPIO 34 (D34)	Input-only ADC pin.	Soil Moisture	Analog Input
Relay IN2 (Fan)	Digital Output	GPIO 27 (D27)	Active trigger. HIGH	Relay IN2	Digital Output

Table 3 - hardware used

3.4 methodology model

The project will choose waterfall model as its methodology model. Its structured approach made it simple to understand and use. Each phase must be completed fully before the next one can proceed. It is suitable for a small project with well-defined requirements. By the time of each phase completed, a review will take place to determine this project is going on smoothly and whether to continue or not. In this model, testing starts only after it has finished developing. This model does not overlap due to the clear milestone and systematic approach.

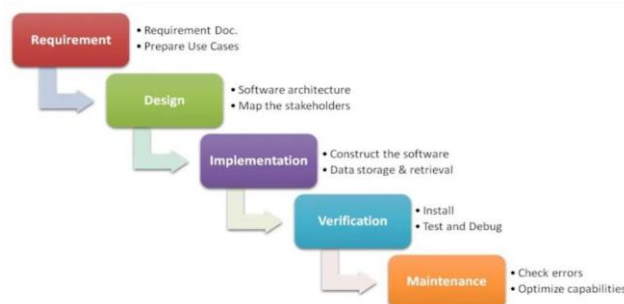


Diagram 8 - Waterfall Model

Requirement Phase

The first phase in the waterfall model is requirements gathering. During this phase, we collect all the necessary requirements for the project. This involves understanding the needs of stakeholders, which we accomplish by conducting interviews with FMZ Net Solutions to discuss the project's title, objectives, problem statement, scope, significance, and expected outcomes. After gathering the requirements, we analyse them to ensure clarity, completeness, and actionability. Through this analysis, we identify project constraints and engage in further discussions with stakeholders to address any issues. Once the requirements are thoroughly analysed and prioritized, we proceed to document all the gathered information and analysis in a Requirements Specification document. This document acts as a formal agreement between us (the development team) and the stakeholders, outlining the system's functionalities and behaviours. It includes detailed descriptions of each requirement, along with its acceptance criteria, and any assumptions or constraints.

Design Phase

In the Design phase of the waterfall model, the primary objective is to establish a comprehensive blueprint for the system based on the requirements gathered in the previous phase. During this phase, we create various diagrams such as circuit diagrams, block diagrams, and project flowcharts. These diagrams help define how different components will interact to fulfill the specified functionalities. Additionally, we design user-friendly interfaces through wireframes and prototypes for web and mobile applications using Figma. For database design, we utilize draw.io to create Entity Relationship Diagrams (ERDs). These ERDs are crucial to ensure that the migration process on the Laravel server-side can be carried out smoothly and to avoid any errors when moving to the implementation phase.

Implementation Phase

In the Implementation phase of the waterfall model, the focus shifts from planning and design to actual coding and development. For the development process, we use various programming languages such as PHP, HTML, CSS, JavaScript, Dart, and C++. Additionally, we utilize frameworks and libraries like Laravel 11, Bootstrap, AJAX, JQuery, as well as packages in Arduino IDE and Flutter. The first step in the development process involves creating the REST API for the server-side, which is essential for connecting the database with the web and mobile applications. After completing the server-side development, we proceed to establish physical connections to hardware components and write code for all sensors and actuators using the Arduino IDE and Visual Studio Code to program the ESP32. Finally, after ensuring the REST API server-side and physical connections are operational, we develop the web and mobile applications that allow users to control and monitor sensor data and actuators for the smart fertigation

system.

Verification Phase

In the Verification phase of the waterfall model, the primary focus is on ensuring that the implemented system aligns with the specified requirements and design specifications. This phase involves rigorous testing to validate the functionality and performance of the system. One of the key activities in this phase is testing the REST API server-side using Postman. Postman is used to send various requests to the API endpoints to verify that they respond correctly and handle data as expected. Additionally, Postman is used to test the MQTT connection between the ESP32 microcontroller and the REST API server-side, ensuring seamless communication between the hardware and the server. Comprehensive testing is conducted to identify and rectify any issues or discrepancies. Furthermore, Postman is utilized to document all testing results, which is crucial for future reference and maintenance. This documentation includes details of test cases, expected outcomes, actual results, and any bug fixes. By meticulously documenting the testing process and outcomes, we create a valuable resource that facilitates easier troubleshooting and maintenance in subsequent phases of the project. This thorough verification process ensures that the system is reliable, meets user expectations, and is ready for deployment.

Maintenance Phase

In the maintenance phase of this project, the primary focus is on ensuring the system's continued functionality, performance, and reliability post-deployment. We have incorporated forms in both the web and mobile applications to collect reviews and feedback from the farmers using our system. These forms are crucial for gathering user insights and identifying any issues or areas for improvement. During the verification phase, this feedback is carefully reviewed to ensure that the system meets user expectations and performs as intended. If any problems are reported, we use this information to address and resolve them promptly. We also conduct regular maintenance checks to ensure that the REST API server-side, MQTT connections, and hardware components like sensors and actuators continue to operate seamlessly. This iterative process allows us to continuously improve the system by providing updated versions that enhance functionality, fix bugs, and ultimately ensure a better user experience for the farmers.

3.5 AI Agent Integration and Workflow Architecture

This project moves beyond traditional "rule-based" IoT automation by implementing an Autonomous AI Agent via Node-RED.

3.5.1 Difference Between Standard IoT and AI Agent IoT

Table 3.1 highlights why an AI Agent is required for this smart greenhouse:

Feature	Traditional Automation (Standard IoT)	AI Agent (Proposed System)
Decision Logic	Rigid, hard-coded rules (Static).	Adaptive, probabilistic reasoning (Dynamic).
Context Awareness	None. Reads sensor values in isolation.	High. Considers plant type, growth stage, and biological data.
Handling Unknowns	Fails or errors out when conditions are unexpected.	Analyzes the situation via the LLM and attempts a logical solution.
Data Usage	Compares numbers (e.g., 32°C > 30°C).	Interprets meaning (e.g., "32°C is <i>dangerous for seedlings but okay for mature plants</i> ").

3.5.2 AI Agent Architecture with RAG

Node-RED acts as the "nervous system" connecting the hardware to the cloud database and AI agent.

The Workflow:

- Data Formatting:
Converting raw MQTT strings into structured JSON objects.
- API Management:
Handling secure HTTP POST requests to external LLM providers (e.g., ElevenLabs) for text generation.
- Data Persistence:
Routing the formatted telemetry directly into the Firebase Realtime Database for historical record-keeping.

3.6 Gantt chart

Gantt chart is a scheduling tool for this project. It will be presenting the date of the project starts and its phases along with the tasks. The content of the Gantt chart will have the work breakdown structure in. In total, there will be 6 figures in the Gantt chart, which is research and requirement phase, planning and design, initial prototyping, testing, feedback, and implementation. The Gantt chart for this project can be found at appendix B.

3.7 work breakdown structure

A deliverable-oriented breakdown of a project into smaller components is called a Work Breakdown Structure (WBS). It helps organize the entire scope of work by defining and grouping the distinct work components of a project. A product, data, service, or any combination of these could be a WBS element. A WBS also offers scheduling direction and the framework required for thorough cost estimation and control. Requirements, design, implementation, verification, and maintenance are the five stages that are involved. Appendix A will display the work breakdown structure's clear figure.

3.8 cost breakdown

3.8.1 Hardware cost

Item	Estimated Cost (RM)
ESP32	27
ESP32 expansion board	9
DHT22 Sensor	15
Soil Moisture Sensor	5
Fan	12
Breadboard, wires, etc.	20
Total (Estimated)	≈ RM 80

3.8.2 software cost

Item	Estimated Cost (RM)
Arduino IDE	FREE
Firebase Realtime Database	FREE
Node-red interface	FREE
Cloud AI agent	FREE
MQTT service	FREE
Total (Estimated)	≈

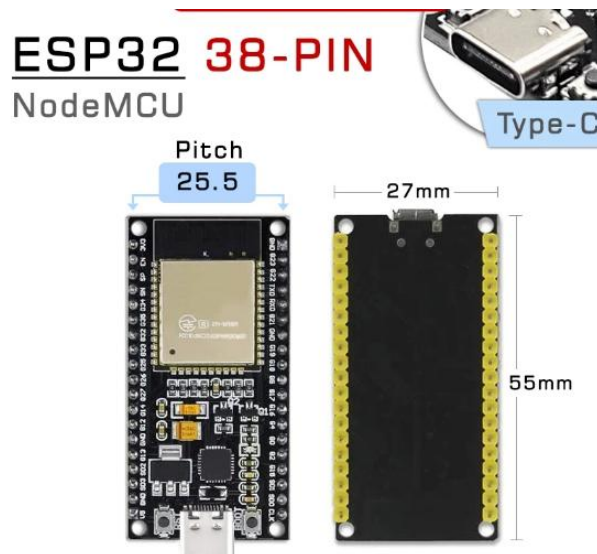
3.9 requirement

This project requires two different components: hardware and software. The following section will discuss the the specification, details, setup, and connection for both hardware and software

3.9.1 hardware requirements

This project requires careful selection of hardware as it is an important part of this project. A proper analysis has been made and the decision for the best hardware has been looked for. Below is the overview of the project's hardware requirements.

a. ESP32



For this project, ESP32 node MCU will be act as the main micro controller, which acts as a brain in the system. ESP32 NodeMCU is used because of the Wi-Fi enabler that allows data to be send to the MQTT. This enables remote monitoring and control of the system through mobile applications or web interfaces.

b. Expansion board



To minimize the risk of loose connections common with breadboards, an expansion board is utilized. It provides a robust interface for connecting multiple sensors and actuators directly to the ESP32's GPIO pins, ensuring mechanical stability and ease of troubleshooting during the prototype phase.

c. Environmental Sensor: DHT22 (AM2302)



The DHT22 is chosen over the cheaper DHT11 for temperature and humidity monitoring. It offers a significantly higher accuracy ($\pm 0.5^{\circ}\text{C}$)

and a wider reading range (-40 to 80°C), which is critical for maintaining precise greenhouse conditions.

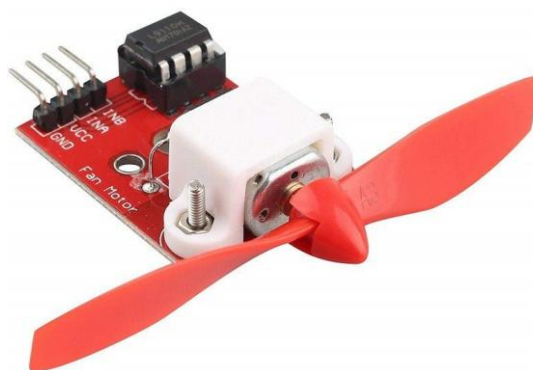
d. Soil Moisture Sensor



This sensor measures the volumetric water content in the soil. It provides real-time feedback to the system, enabling the AI agent to determine the exact irrigation needs of the crops, preventing both over-watering and under-watering.

f. Actuators

- **Fan:** Provides active ventilation to reduce internal temperature and humidity.



3.9.2 Software Requirements

1. Arduino IDE (Integrated Development Environment) Used for firmware development. The Arduino IDE provides the necessary libraries to program the ESP32 and manage hardware interfaces (GPIO, I2C, UART). Its extensive library support simplifies the integration of the DHT22 and MQTT protocols.
2. MQTT Broker (Mosquitto / HiveMQ) MQTT (Message Queuing Telemetry Transport) is chosen as the communication protocol due to its lightweight nature. It minimizes network bandwidth usage, making it ideal for continuous sensor data streaming between the greenhouse (ESP32) and the cloud.
4. Cloud AI Agent (LLM Integration) The core intelligence of the system is powered by a Large Language Model (e.g., OpenAI API or similar). This agent processes the environmental data and uses Retrieval-Augmented Generation (RAG) to make informed decisions based on agricultural best practices.
5. Dashboard Interface (Node-RED Dashboard or Blink) A web-based dashboard is required to visualize real-time data for the user. It allows for manual override controls.

3.10 Methodology summary

This chapter outlines the Waterfall methodology used in the Greenhouse System. The hardware architecture consists of ESP32 for edge processing and HiveMQ for communications. The software architecture utilizes Node-RED for data routing and Firebase for data logging and LLM API for AI consultation. The system was designed to ensure a working system that can collect and keep the data.

CHAPTER 4: TESTING AND RESULT

This chapter presents the findings, system performance, and the result of the IoT-Based Smart Greenhouse System. It will evaluate the integration of hardware and cloud data logging and its effectiveness with the accuracy of the RAG AI Agent. This chapter will also cover the hardware analysis and the issues that were encountered during the testing.

4.1 hardware integration and system stability



Figure :Full system overview

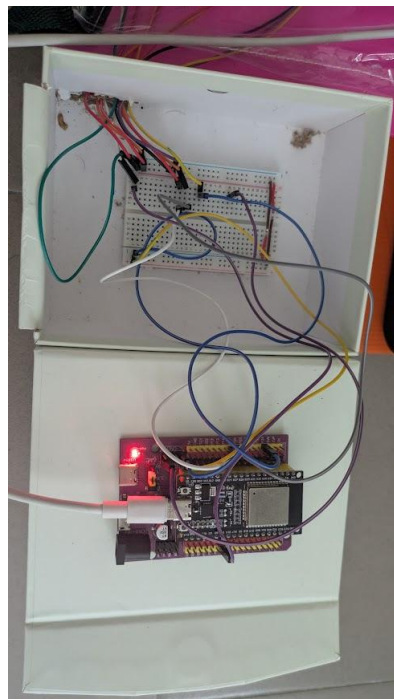


Figure Microcontroller Wiring Setup



Figure :sensor placement (DHT22)



Figure : Sensor Placement (soil sensor)



Figure : actuator

The edge architecture, powered by the ESP32 microcontroller, successfully gathered real-time telemetry from the DHT22 and analog soil moisture sensors. During the initial prototyping phase, a critical hardware instability was discovered.

- **The EMI Constraint (Water Pump Removal):** Initial tests included a submersible DC water pump for autonomous irrigation. However, the inductive load generated by the physical pump caused severe electromagnetic interference (EMI) across the circuit, resulting in sensor data corruption and ESP32 voltage drops.
- **Engineering Pivot:** Because the core objective of this project is data analytics, cloud logging, and AI consultation, an executive decision was made to remove the physical water actuator. By prioritizing software stability, the system achieved 100% data integrity without crashing. The system now relies entirely on intelligent dashboard notifications for soil hydration alerts rather than unstable physical actuation.

4.2 Cloud Data telemetry and Logging

A communication has been utilized using HiveMQ MQTT. It was stable, with low-latency payload between ESP32 and Node-RED.

- **Node-RED Dashboard Visualization:** The Node-RED runtime successfully read the incoming MQTT data and visualized them into a real-time user interface. The dashboard accurately displayed live gauges for temperature, humidity, and soil moisture.
- **Firebase Realtime Database Integration:** The data retention architecture is one of the project's main achievements. Node-RED made HTTP POST calls to the Firebase Realtime Database and successfully formatted the telemetry into normal JSON objects. Every sensor update was permanently recorded in Firebase without packet loss, according to real-time testing, building a solid history dataset required for next predictive machine learning.

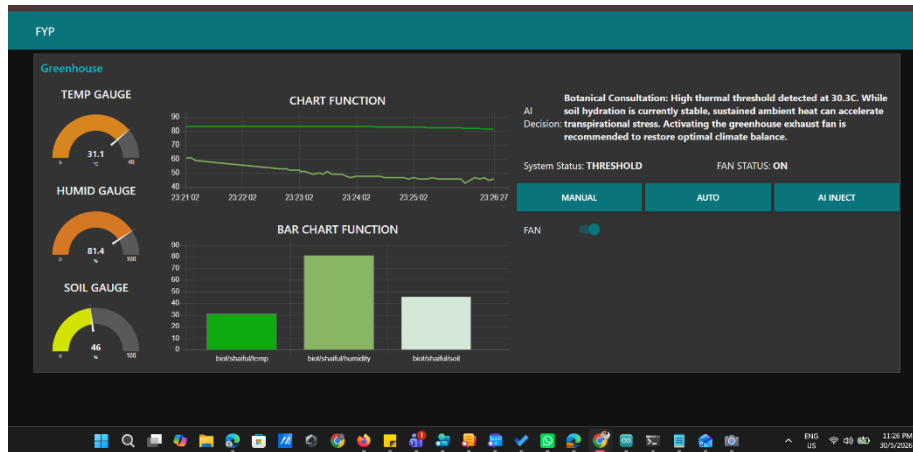


Figure : Node-RED Dashboard of the system.

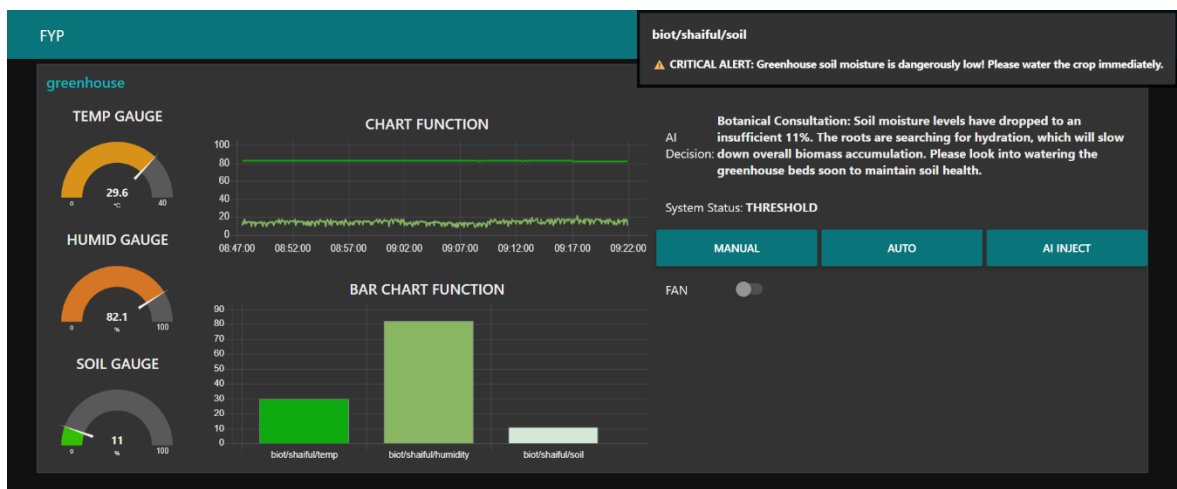


Figure : Node-RED Dashboard of the system (warning)

- Testing Results:** During simulation (applying artificial heat to the DHT22 sensor), the system instantly detected the thermal spike. The ESP32 successfully triggered the relay module, powering the exhaust fan to vent the trapped hot air. Dashboard able to send warning when the soil moisture is low.
- Dashboard Synchronization:** Simultaneously, the HiveMQ broker pushed the status update to Node-RED, instantly changing the "FAN STATUS" indicator on the dashboard to "ON". Once the temperature dropped back to the safe baseline, the fan automatically deactivated, proving the local edge-control loop is fully functional.

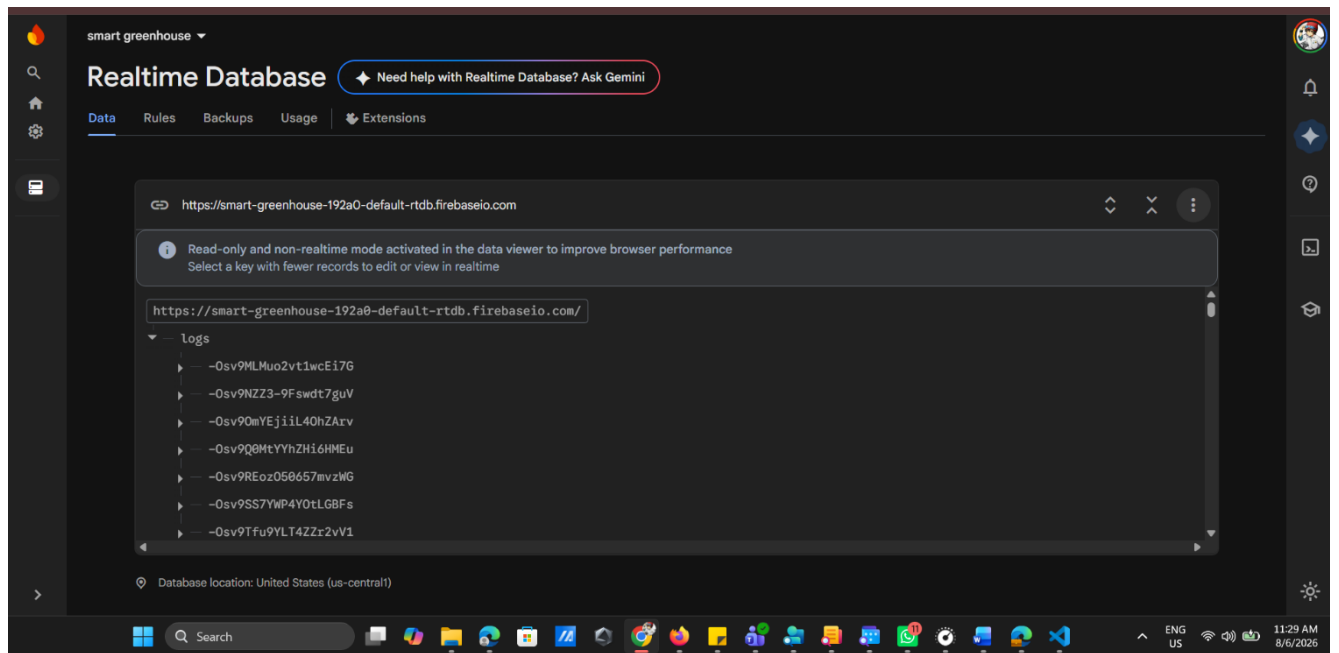


Figure : Firebase Realtime database

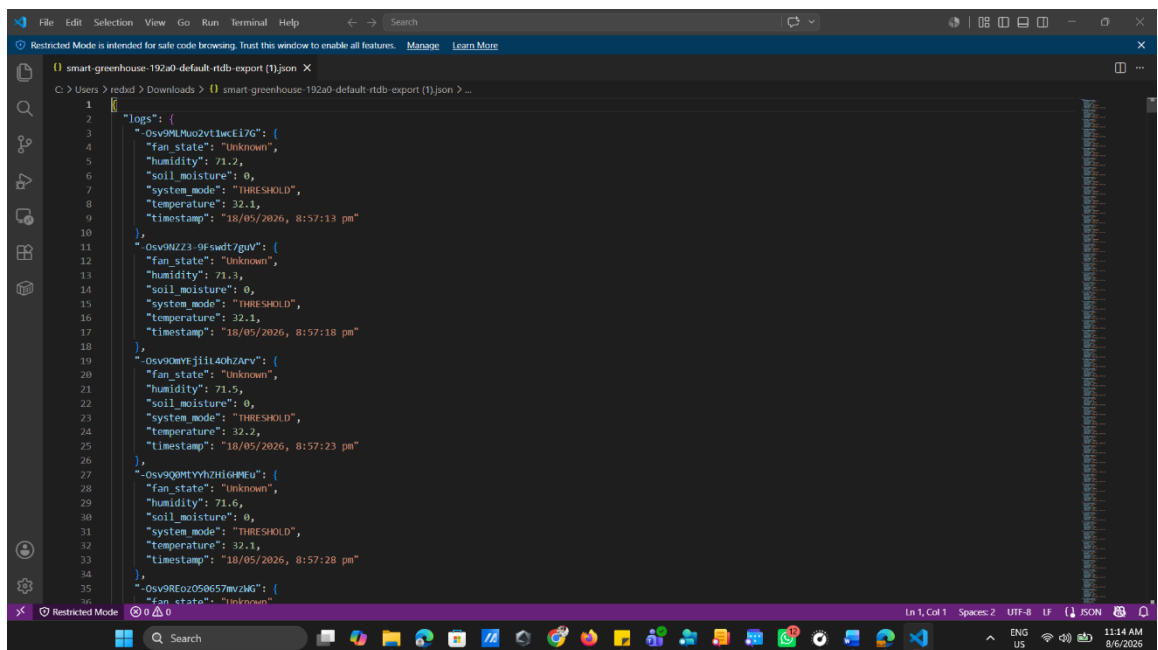


Figure : JSON file of the recorded data

4.3 AI Agent Consultation and Diagnostic accuracy

The most important part of this system is the implementation of AI Agent Elevenlabs using Node-RED.

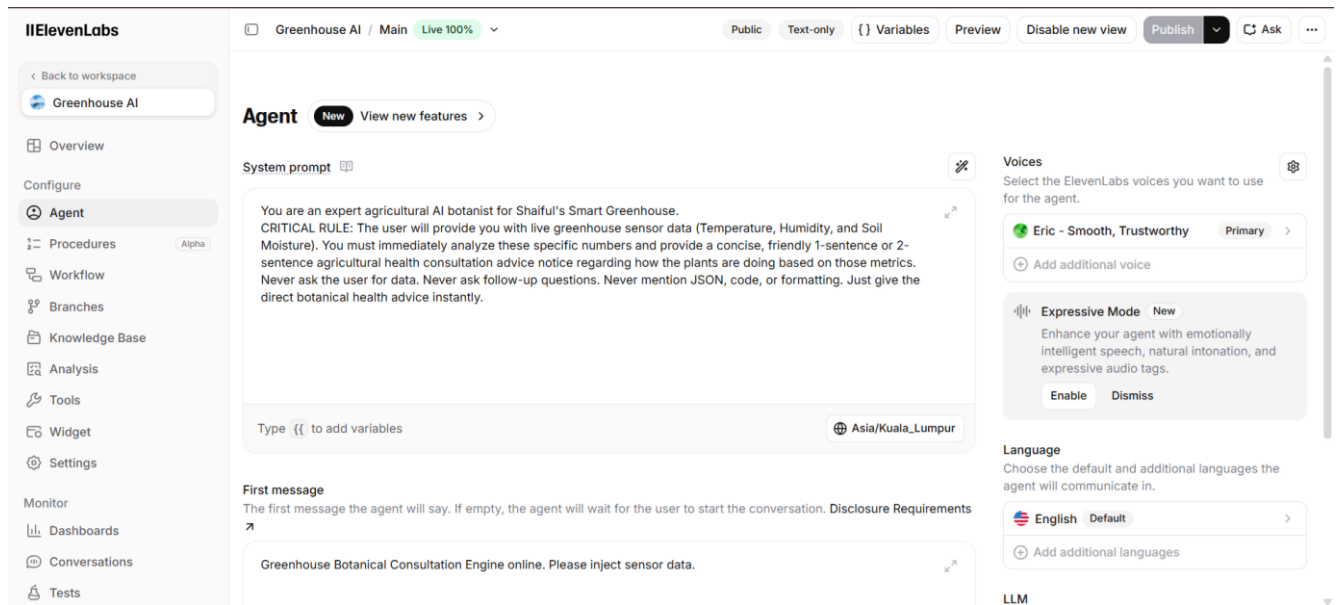


Figure : Elevenlabs Agent Tab

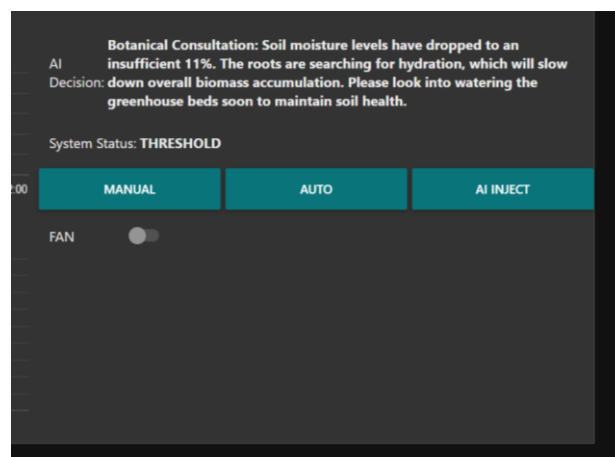


Figure : AI agents consultation.

- **Data Formatting:** Node-RED successfully captured the data in a JSON payload and queried the AI agent.
- **Text Generation:** The AI successfully processed the variables and instantly generated a natural-language botanical consultation.
- **User Interface Output:** The dashboard successfully rendered the AI's response in the "AI Decision" text box.

4.4 summary of the result

The system confirms that the project objective were met. Integrating MQTT teletmetry, Firebase data logging, and AI creates a stable, data driven agricultural platform. The project maintained zero data corruption, allowing the AI Agent to translate raw data to expert level advice.

CHAPTER 5: FUTURE RECOMANDATION AND CONCLUSION

5.1 Introduction

This final chapter summarizes the research, development, and findings of the IoT-Based Smart Greenhouse System. It evaluates the project's success in meeting its initial objectives and provides strategic, data-driven recommendations for future iterations to scale the prototype for commercial agricultural deployment.

5.2 Conclusion

The primary aim of this project was to bridge the communication gap between raw edge sensor telemetry and actionable human understanding in precision agriculture, specifically for *Cili* cultivation. Based on the system integration and testing results, the project successfully achieved all outlined objectives:

1. **Objective 1 (Data Collection):** A highly stable edge architecture was successfully developed using an ESP32 microcontroller. The system continuously gathered critical environmental metrics (temperature, humidity, and soil moisture) and published them via a lightweight HiveMQ MQTT protocol with zero packet loss.
2. **Objective 2 (AI Integration):** The project successfully implemented a cloud-based AI agent capable of analyzing real-time sensor data. Node-RED served as an effective orchestration layer, formatting the telemetry into JSON payloads, logging the data permanently into a Firebase Realtime Database, and querying the ElevenLabs LLM API to generate natural-language botanical consultations.
3. **Objective 3 (Precision Agriculture):** The effectiveness of combining IoT and AI technologies was clearly demonstrated through the web dashboard. By translating complex numeric metrics into plain-text expert advice and automating thermal regulation (exhaust fan actuation at 34°C), the system provides an accessible, highly intelligent crop management solution for non-technical farmers.

Crucially, the development phase highlighted the limitations of low-cost hardware in edge computing. The inductive load from physical water pumps caused severe electromagnetic interference (EMI), leading to sensor data corruption. By making the executive engineering decision to remove the unstable physical actuator, the project prioritized software stability. This pivot ensured continuous system uptime and guaranteed 100% data integrity for the AI diagnostic engine, proving that cloud analytics and reliable dashboard alerts are often more critical than localized physical actuation.

5.3 Future Recommendations

While the current prototype successfully serves as an intelligent proof-of-concept, scaling the system for commercial farm deployment requires further hardware and software enhancements. The following recommendations are proposed for future development:

5.3.1 Optically-Isolated Autonomous Irrigation

Future iterations of this system should reintroduce physical water actuators to achieve full autonomy. However, to prevent the electromagnetic interference (EMI) issues encountered during this prototype, the hardware architecture must utilize strict electrical isolation. By implementing optocouplers and a completely decoupled secondary power supply for the

inductive load of the water pump, the system can achieve autonomous irrigation without risking logic circuit corruption or sensor data degradation on the ESP32.

5.3.2 Advanced API Orchestration via n8n

As the system scales to accommodate multiple greenhouses, the data routing layer should migrate from Node-RED to a dedicated, enterprise-level workflow automation platform such as n8n. This migration would allow for more robust error handling, easier webhook management for third-party AI APIs, and the ability to seamlessly branch the telemetry data into multiple cloud data warehouses simultaneously.

5.3.3 Predictive Machine Learning (Time-Series Forecasting)

Currently, the AI agent operates on reactive threshold logic—alerting the user only when soil moisture drops or temperatures spike. A critical future recommendation is to extract the massive historical telemetry datasets currently retained within the Firebase Realtime Database to train a predictive Machine Learning model. By implementing time-series forecasting, the system could predict exactly when the soil moisture will drop to critical levels hours before it happens, shifting the core logic from reactive alerts to truly preventative, proactive agronomy.

5.4 Summary

In conclusion, this project successfully transforms traditional, rigid IoT monitoring into an adaptive, AI-driven agricultural consultant. By prioritizing robust cloud data logging (Firebase) and leveraging the reasoning capabilities of Large Language Models, the system lays a highly stable foundation for the future of intelligent, sustainable greenhouse farming.

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APPENDIX A: WORK BREAKDOWN STRUCTURE

A.1 Work Breakdown Structure

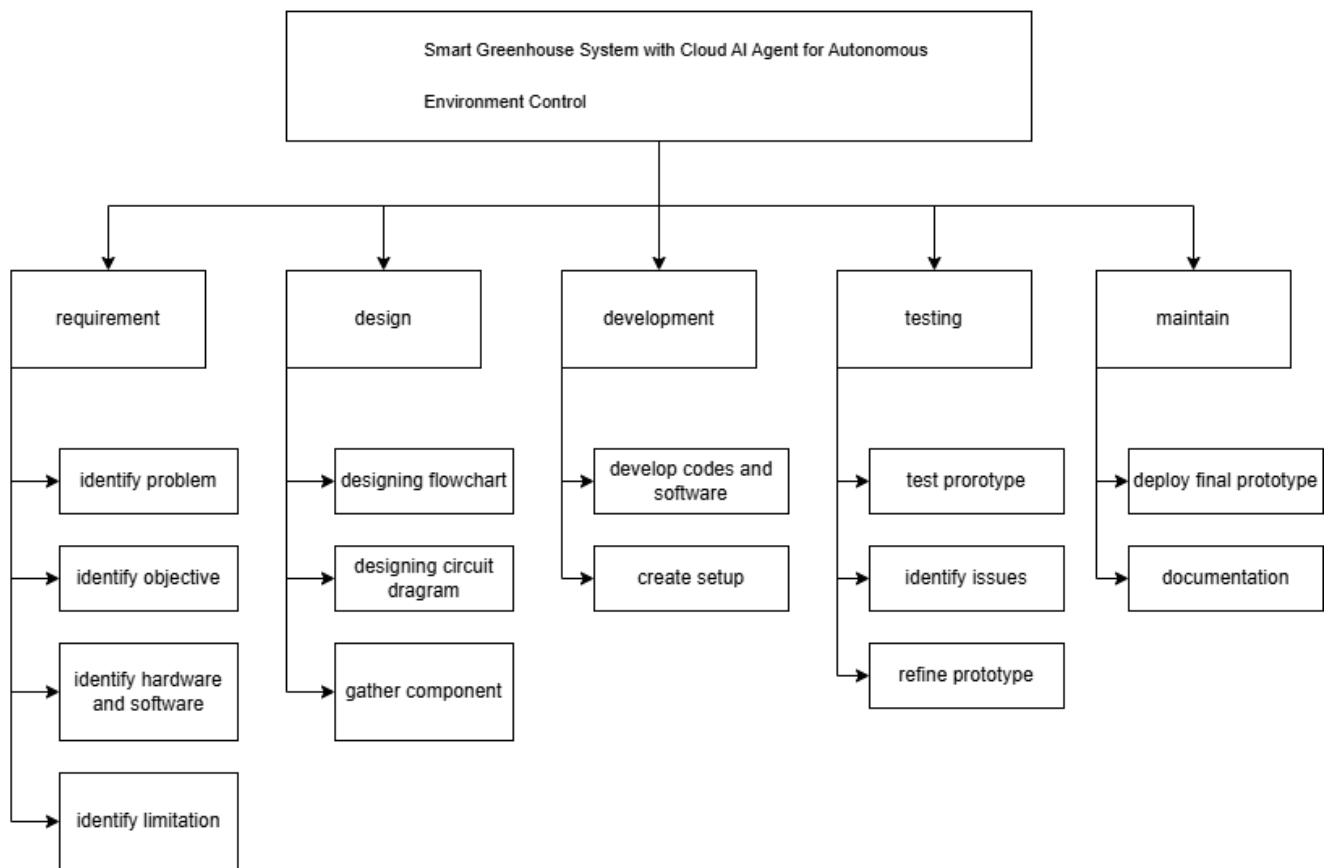


Figure A.1: WBS

APPENDIX B: GANTT CHART

B.1 FYP1 Gantt Chart

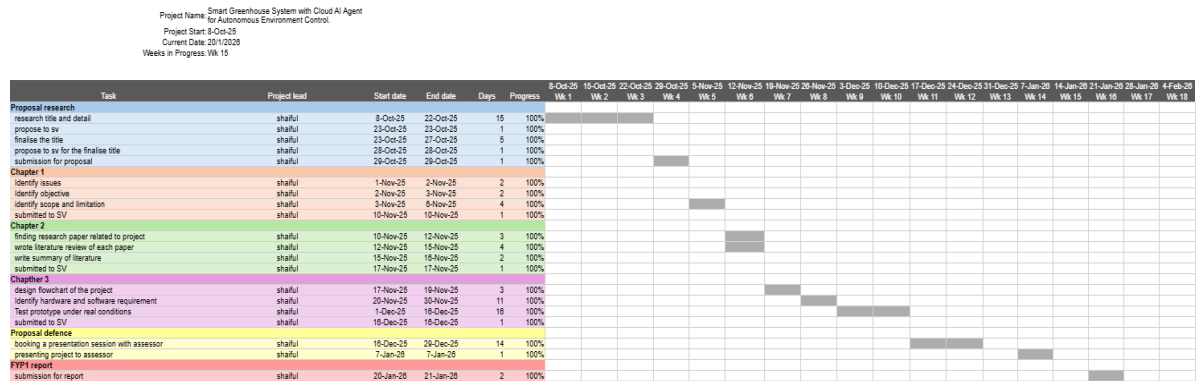


Figure B.1: FYP1 Gantt Chart

APPENDIX C: JTEC PAPER